Classifying AI-Generated vs Real Images using CNNs

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Problem Statement

The aim of this project is to build an image classification model that can distinguish between **AI-generated images** and **real images**. With modern generative models producing highly realistic images, it has become difficult to identify fake visual content.

This project formulates the task as a binary classification problem:

- Real Image
- AI-Generated Image

Program Input and Output

Input: RGB image (resized to a fixed size, e.g., 128×128)

Output: Predicted label (Real / AI-generated) with an optional confidence score

Data Sources

The dataset will be taken from Kaggle:

- **Link:** `https://www.kaggle.com/datasets/cashbowman/ai-generated-images-vs-real-images/data`
- Data type: Image files (JPG/PNG)
- Labels: Provided (Real vs AI-generated)

The data will be split into training, validation, and test sets.

Approach

A Convolutional Neural Network (CNN) will be used for classification. CNNs are suitable because they can learn visual features such as textures, edges, and patterns directly from images.

CNNs can help detect:

- Texture irregularities
- Lighting inconsistencies
- Repeated or artificial patterns
- Smoothing or generation artifacts

Feasibility

This project is feasible within 3 weeks since the dataset is readily available and the task is a simple binary classification problem.

Timeline:

- Week 1: Data preprocessing and model setup
- Week 2: Training and validation
- Week 3: Testing and analysis

Proposed Goals and Success Criteria

The main goals of the project are:

- Prepare and pre process the dataset
- Train a CNN model for classification
- Evaluate using accuracy and related metrics
- Test on unseen images
- Analyze where the model fails

Success criteria:

- The model successfully trains and converges
- Reasonable classification performance on test data
- Clear and interpretable predictions

Expected Outcome

The expected outcome is a functional model capable of classifying images as AI-generated or real. The final output will include:

- Trained model
- Evaluation metrics and plots
- Example predictions
- Discussion of limitations and improvements.